Attachment D

Affidavit of Danielle Croop

UNITED STATES OF AMERICA
BEFORE THE
FEDERAL ENERGY REGULATORY COMMISSION

PJM Interconnection, L.L.C.) Docket No. ER27-____-000

AFFIDAVIT OF DANIELLE CROOP
ON BEHALF OF PJM INTERCONNECTION, L.L.C.

1. My name is Danielle Croop, Senior Manager in the Market Design group for PJM Interconnection, L.L.C. ("PJM"). My business address is 2750 Monroe Blvd., Audubon, Pennsylvania, 19403. I am submitting this affidavit on behalf of PJM to support the proposed reforms to PJM's Open Access Transmission Tariff ("Tariff") and PJM's Reliability Assurance Agreement Among Load Serving Entities ("RAA") to implement the Reliability Backstop Procurement ("RBP") framework by providing support for PJM's proposal to:

    a. Update the Initial RBP Target; and
    b. Update Cost Allocation Methodology to Electric Distributor zone/areas.

Qualifications

2. In my role as Senior Manager for PJM, I lead a team of Market Design Specialists and Economists on the development of PJM's market design across all of PJM's markets including the energy, ancillary services, and capacity markets. My responsibilities include leading the development of wholesale market rules and participation frameworks that advance market efficiency and grid reliability. I joined PJM in 2014 as an Engineer, and since then, I have held numerous other roles, including as a Senior Engineer, Senior Lead Engineer, Senior Lead Market Design Specialist, and Market Design Manager, providing technical support and leading cross-divisional teams on market design, market evolution initiatives related to energy storage and distributed energy technologies participating in the PJM markets, and balancing operations at PJM, including involvement in a number of North American Electric Reliability Corporation subcommittees and initiatives.

3. Prior to joining PJM, I was employed by the SI Organization, Inc. as a System Engineer supporting large system integrations. I received a Bachelor of Engineering Degree and a Master of Engineering (Electrical Engineering) from Drexel University in Philadelphia, Pennsylvania, and have obtained a Certificate in Engineering Management.

Initial RBP Target

4. PJM proposes to update the Initial RBP Target using the lesser of the projected shortfall established from the latest load forecast and planning parameters for the 2028/2029 Delivery Year that will be available at the time the RBP is conducted, and the aggregate quantity of Large Load growth reported to PJM, rather than

relying solely on the shortfall quantity established through the Base Residual Auction for the 2028/2029 Delivery Year, held in July 2026.

5. First, utilizing the latest information allows PJM to identify the most accurate estimate of the reliability need for the 2028/2029 Delivery Year based on the latest load forecast of demand that will need to be served and the expected supply. Specifically, because this updated information will be available prior to conducting the RBP, it is appropriate to use the most up-to-date information to produce an Initial RBP Target that represents the latest expected reliability shortfall and avoid either under-procurement or over-procurement of resources.

6. Second, applying the "lesser of" calculation using the aggregate quantity of Large Load growth reported to PJM for the RBP will mitigate the potential over-allocation of RBP Charges to a zone/area relative to the forecasted Large Load growth. PJM believes this scenario is unlikely based on currently available information. Specifically, PJM's 2026 Load Forecast indicates approximately 9,000 MW of large load growth, while the projected Regional Transmission Organization ("RTO") shortfall for the 2028/2029 Base Residual Auction is approximately 6,800 MW. Although these values may change as updated forecasts become available in 2027, PJM expects the relationship between Large Load growth and the projected shortfall to remain generally consistent. Nevertheless, including the "lesser of" calculation in the Initial Target provides an important safeguard by accounting for the possibility that previously reported Large Load agreements may be canceled between the collection of data for the 2027 Load Forecast and the establishment of the RBP. By ensuring that RBP cost allocations do not exceed 100% of the Large Load growth reported for the applicable zone/area, the proposed procurement target and cost allocation framework better align cost responsibility with the principles of cost causation and beneficiary pays.

7. Further, since PJM plans to run the RBP in March 2027, by that time PJM will have conducted the 2029/2030 Base Residual Auction, scheduled for December 2026. That auction may produce a shortfall, and of different quantity than for the 2028/2029 Delivery Year. Nevertheless, PJM proposes to maintain the 2028/2029 Delivery Year shortfall as the basis for the RBP. This approach is consistent with the underlying intent of the RBP to address the nearer-term reliability need while allowing market fundamentals to address the longer-term resource adequacy.

8. Additionally, adopting a different procurement target timeframe would necessitate broader revisions to the overall design, including resource eligibility requirements and commitment terms. Because market participants have already planned and made business decisions based on PJM conducting the RBP for the 2028/2029 Delivery Year, maintaining the existing design is prudent. Doing so promotes certainty, preserves stakeholder expectations, and avoids unnecessary disruption to the market.

9. The RBP is intended to serve as a bridge mechanism to address immediate reliability concerns while PJM and stakeholders continue to pursue longer-term

market design reforms aimed at improving investment and retention signals to support resource adequacy. In support of those efforts, PJM has initiated its Powering Reliability Through Market Design initiative with stakeholders and continues to work with the Charles River Associates to facilitate bilateral matching efforts.

10. The shortfall MW calculation that will set the Initial RBP Target will be calculated using the updated information from the 2027 Load Forecast and associated updated planning parameters and submitted Large Load MW values from the Electric Distributors. More specifically, PJM will calculate the Initial RBP Target MW as:

The lesser of:

(1) the PJM Region Reliability Requirement for the 2028/2029 Delivery Year minus the UCAP-equivalent of the ICAP quantity of Capacity Resources that cleared in the 2028/2029 Base Residual Auction

Where the PJM Region Reliability Requirement for the 2028/2029 Delivery Year will be calculated as the RTO Region Peak Load Forecast for the 2028/2029 Delivery Year from the 2027 Load Forecast times the 2028/2029 Forecast Pool Requirement from the updated planning parameters published in 2027 prior to the commencement of the RBP, less the sum of all updated Unforced Capacity Obligations of FRR Entities in the PJM Region.

And where the UCAP-equivalent of the ICAP quantity of Capacity Resources that cleared in the 2028/2029 Base Residual Auction, where the UCAP-equivalent values will be determined by the Capacity Resources' updated capacity accreditation factors for the 2028/2029 Delivery Year that will be available in the updated planning parameters available just prior to the commencement of the RBP.

And

(2) the UCAP equivalent of the aggregate region-wide Large Load MW quantity reported by the Electric Distributor

Where the UCAP equivalent of the aggregate region-wide Large Load MW is the aggregate region-wide Large Load MW times the most recent 2028/2029 Forecast Pool Requirement.

11. The RBP will function as a safety net to address the shortfall MW for the 2028/2029 Delivery Year that has not been addressed by new Large Load bringing their own new capacity through bilateral agreements or otherwise. The RBP is a stop-gap, one-time measure aimed specifically at addressing the resource adequacy shortfall in the PJM Region, an effort designed to work in part with the load that has voluntarily brought supply to the PJM Region—independently or with support of ongoing PJM bilateral matching efforts—and a return to market fundamentals to support the expected load growth over the next decade.

RBP Cost Allocation Methodology

12. The cost allocation structure begins with the Initial RBP Target based on the Large Load growth between the 2026 summer peak load and the forecasted Large Load MW in each Electric Distributor's zone/area for the 2028/2029 Delivery Year. The Initial RBP Target is then adjusted by the quantity of Qualifying RBP Offset UCAP MW and Peak Shaving Adjustment opt-out to reach the Final RBP Target. Each Electric Distributor's zones/areas with resulting forecasted Large Load growth between 2026 and the 2028/2029 Deliver Year will be allocated a pro rata share of the Final RBP Target.[1] The MW quantity assigned to each Electric Distributor's zone/area will be further allocated to Load Serving Entities ("LSE") based on the RBP Obligation ("RBO"), which permits states to determine the appropriate cost allocation to specific LSEs. Should the Relevant Electric Retail Regulatory Authority(s), through the relevant Electric Distributors, decline to provide an RBO MW, PJM has proposed to apply a backup default cost allocation where costs would be allocated to all of the LSEs in the zone/area based on their percent share of the zone/area's Daily Unforced Capacity Obligation.

Step 1: Collecting the Forecasted Large Load Growth Between 2026 Through the 2028/2029 Delivery Year Subject to RBP Cost Allocation

13. Seven days prior to the commencement of the RBP, PJM will receive all Large Load growth in the Electric Distributor zone/area from the 2026/2027 Delivery Year through the 2028/2029 Delivery Year. The Large Load growth will be submitted based on firm commitments, as supported by electric service obligation, construction commitment, or equivalent agreements, and measured based on summer peak values. These submissions are required to align with the Large Loads submitted to PJM for the 2027 load forecast. Any discrepancies between the Electric Distributors' submissions for RBP and data submitted to PJM for the 2027 load forecast must be accompanied by written documentation of the change from the Electric Distributor.

14. The aggregate Large Loads growth will be submitted within a zone/area, along with identification of the underlying Large Loads. This will result in PJM having the aggregate value to determine the total Large Load MW for the zone/area and the summation of all zone/areas to determine the total Large Load MW for the RTO Region. The underlying Large Loads will be used to facilitate the Qualifying RBP Offset UCAP MW process.

Step 2: Allocating to Electric Distributor Zone/Area

15. To allocate costs, i.e., RBP Charges, PJM is proposing to determine cost responsibility by the Electric Distributor zone/area and their pro rata share of Large Load MW subject to RBP in comparison to the total Large Load MW subject to

[1] This cost allocation structure in detailed in the example below starting in paragraph 17.

RBP across the RTO. The final pro rata share determination will be fixed for the term of the RBP and be applied to the committed RBP UCAP MW and associated RBP Charges throughout the term of the RBP.

16. An illustrative example of Step 1 and Step 2 is detailed in the following section.

Example of Allocation to Electric Distributor Zone/Area

17. For this example, the Initial RBP Target MW will be assumed to be 4,000 MW. This value is purely illustrative for this example and not indicative of the Initial RBP Target for the RBP.

18. The initial pro rata allocation to each Electric Distributor zone/area will be based on the submitted data from the Electric Distributors representing the forecasted Large Load growth between the 2026/2027 Delivery Year and 2028/2029 Delivery Year. The initial share is calculated as: [Zone/Areas Large Load Growth MW/ RTO Large Load Growth MW], which can be represented as the initial RBP UCAP MW allocation based on that pro rata share of the Initial RBP Target.

Zone/Area	Submitted Large Load Growth (MW)	Initial Share of RBP (%)	Share of Initial Target MW (4000MW)
Zone/Area 1	1000	24%	975.6
Zone/Area 2	500	12%	487.8
Zone/Area 3	100	2%	97.6
Zone/Area 4	1000	24%	975.6
Zone/Area 5	1500	37%	1463.4
Total	***4100***	***100%***	***4000***

19. Second, as Qualifying RBP Offset UCAP MW are submitted, both the Final RBP Target and the pro rata allocation to each Electric Distributor zone/area will be updated. In this example, I will assume the following Qualifying RBP Offset UCAP MW were submitted and will decrement the Initial RBP Target:

Zone/Area	Load (MW)	Qualifying RBP Offset UCAP MW
Zone/Area 1	500	470
Zone/Area 2	100	94

Note: The Qualifying RBP Offset UCAP MW is represented by Load*FPR. FPR is represented as 0.9401 int this example.

20. In my illustrative example, for purposes of simplicity, I assume there are no additional decrements to the Initial RBP Target for opt outs. Thus, the Final RBP Target would be calculated as Initial RBP Target – Qualifying RBP Offset UCAP

MW. In this example, this would be: 4,000 MW – [470 MW + 94 MW] = 3,436 MW.

21. Subsequently, the Large Load that submitted the Qualifying RBP UCAP MW will be removed from the Large Load growth for cost allocation purposes. The updated pro rata allocation and RBP UCAP share of the Final RBP Target is demonstrated in the table below. In Zone/Area 1 the Large Load Growth (MW) will be reduced from 1000 MW to 500 MW to account for the 500 MW of load with demonstrated Qualifying RBP Offset UCAP MW. Similarly, Zone/Area 2 will be updated to 400 MW to account for the Qualifying RBP Offset UCAP MW.

22. Based on adjustments to each Electric Distributor zone/area's Large Load Growth MW, the final pro rata shares are recalculated.

Zone/Area	Submitted Large Load Growth (MW)	Final Share of RBP (%)	Share of Final Target MW (3436 MW)
Zone/Area 1	500	14%	490.9
Zone/Area 2	400	11%	392.7
Zone/Area 3	100	3%	98.2
Zone/Area 4	1000	29%	981.7
Zone/Area 5	1500	43%	1472.6
Total	***3500***	***100%***	***3436***

The Bounds of Qualifying Offsets

23. Large Load can be removed from the Large Load MW used to determine RBP cost allocation by bringing Qualifying RBP Offset UCAP MW. A Large Load can bring up to their capacity obligation (in UCAP), determined as peak MW times the Forecast Pool Requirement. MW from supply resources in excess of that value will not be considered to decrement the Large Load growth in a zone/area or decrement the Initial RBP Target value.

24. A zone/area cannot reduce its zone/area share of the Initial RBP Target beyond the capacity obligation of the total Large Load MW in that zone/area (floored at zero). Further, a zone/area must provide Qualifying RBP Offset UCAP MW equal to capacity obligation of the total Large Load MW in that zone/area to be fully removed from the RBP cost allocation.

25. This concludes my affidavit.

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VERIFICATION

I, Danielle Croop, pursuant to 28 U.S.C. § 1746, state, under penalty of perjury, that I am the Danielle Croop referred to in the foregoing document entitled "Affidavit of Danielle Croop, Senior Manager in the Market Design group for PJM Interconnection, L.L.C.," that I have read the same and am familiar with the contents thereof, and that the facts set forth therein are true and correct to the best of my knowledge, information, and belief.

/s/ Danielle Croop
Danielle Croop
Sr. Manager, Market Design
PJM Interconnection, L.L.C.